

2/11/2026

instead of α & $\beta \rightarrow A_e$ and phase transformation δ_e

$$\text{for 2) } A_e [\cos \delta_e J_e(Kr) - \sin \delta_e N_e(Kr)]$$

$$r \rightarrow \infty \quad R_{ke}(r) \sim \frac{A_e}{Kr} \sin(Kr - \pi/2 + \delta_e)$$

at $r=a \rightarrow R_{ke}(r)$, $r < a = R_{ke}(r)$, $r > a$

That's the main idea behind this.

Boundary Conditions at $r=a$

$$\frac{\psi'}{\psi} \text{ continuous}$$

$$\delta_e = \frac{x J_e'(xa)}{J_e(xa)}$$

$$\text{where } J_e'(x) = \frac{dJ_e(x)}{dx}$$

$$= K \frac{J_e'(ka) \cos(\delta_e) - N_e'(ka) \sin(\delta_e)}{J_e(ka) \cos(\delta_e) - N_e(ka) \sin(\delta_e)}$$

$$\delta_e \cdot \left[J_e(ka) \frac{\cos \delta_e}{\cos(\delta_e)} - N_e(ka) \frac{\sin(\delta_e)}{\cos(\delta_e)} \right] = K \left[J_e'(ka) \frac{\cos(\delta_e)}{\cos(\delta_e)} - N_e'(ka) \frac{\sin \delta_e}{\cos \delta_e} \right]$$

$$\rightarrow \tan(\delta_e) = \frac{-K J_e'(ka) - \delta_e J_e(ka)}{K N_e'(ka) - \delta_e N_e(ka)}$$

$$\delta_e = 0$$

$$ka \ll 1$$

$$\lambda \gg a$$

$$ka = x$$

$$J_0(x) = \frac{\sin(x)}{x}, \quad J_0'(x) = \frac{\cos(x)}{x} - \frac{\sin(x)}{x^2}$$

$$N_0(x) = -\frac{\cos(x)}{x}, \quad N_0'(x) = \frac{\sin(x)}{x} + \frac{\cos(x)}{x^2}$$

$$\tan(\delta_0) = \frac{x J_0'(x) - a \gamma_0 J_0(x)}{x N_0'(x) - a \gamma_0 N_0(x)} = \frac{x \left[\frac{\cos x}{x} - \frac{\sin x}{x^2} \right] - a \gamma_0 \frac{\sin x}{x}}{x \left[\frac{\sin x}{x} + \frac{\cos x}{x^2} \right] + a \gamma_0 \frac{\cos x}{x}}$$

$$x \rightarrow 0$$

$$\Rightarrow \tan(\delta_0) \approx \frac{-a \gamma_0}{1 + a \gamma_0} x = \frac{-a \gamma_0}{1 + a \gamma_0} (ka)$$

ka might be small but that does not mean δ_e is going to be.

We need to now calculate $a \gamma_0$...

"a good scandal is always good right?" - Yvrii

$$a\gamma_0 = \frac{ax \int_0(xa)}{\int_0(xa)} = \frac{xa \left(\frac{\cos xa}{xa} - \frac{\sin xa}{(xa)^2} \right)}{\sin xa / xa}$$

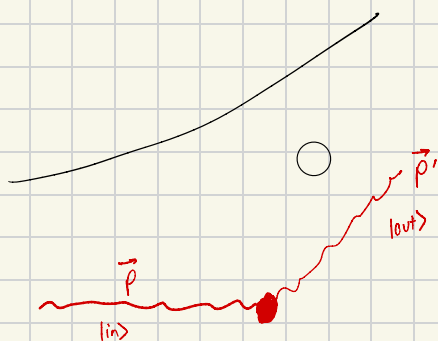
$$= \frac{xa \cot(xa) - 1}{}$$

$$\rightarrow \frac{-a\gamma_0}{1+a\gamma_0} (ka) = (-ka) \frac{xa \cot(xa) - 1}{xa \cot(xa)}$$

$$\rightarrow -ka \left[1 - \frac{\tan xa}{xa} \right]$$

$$\rightarrow \sigma_0 = \frac{4\pi}{k^2} \sin^2[\delta_0] = \frac{4\pi a^2}{} \left[1 - \frac{\tan xa}{xa} \right]^2$$

now impact parameter size $\gg$ size of scattering object



We are treating the potential like a perturbation.

BORN APPROXIMATION

quite a universal approximation

Fermi Golden Rule

Probability of initial to final state

$$|in\rangle \quad t \rightarrow -\infty$$

$$|out\rangle \quad t \rightarrow +\infty$$

$$dW_{fi} = \frac{2\pi}{\hbar} |V_{fi}|^2 \delta(E_f - E_i) dz_f$$

Energy conserves

State density for a final state

$$dz_f = \frac{d^3 \vec{p}'}{(2\pi\hbar)^3}$$

matrix element of potential of initial and final states

$$V_{fi} = \langle out | \hat{U} | in \rangle_f$$

$$S(f(x)) = \sum \frac{S(f(x_i))}{|P'(x_i)|}$$

x_i : zeros of $f(x)$

Talking about the NON-RELATIVISTIC CASE

$$\delta(E_f - E_i) = \delta(p'^2/2m - p^2/2m) = 2m \delta(p'^2 - p^2)$$

So now initial state p , final state $p' \rightarrow$

$$dW_{p \rightarrow p'} = \frac{4\pi m}{\hbar} |V_{\vec{p}' \vec{p}}|^2 \delta(p'^2 - p^2) \frac{d^3 \vec{p}'}{(2\pi\hbar)^3}$$

Normalization of $|\psi\rangle \rightarrow$

1 particle / second / area unit

$$\int \vec{j} \cdot d\vec{A} = 1$$

$\vec{j} \rightarrow$ probability current
 $A \rightarrow$ area